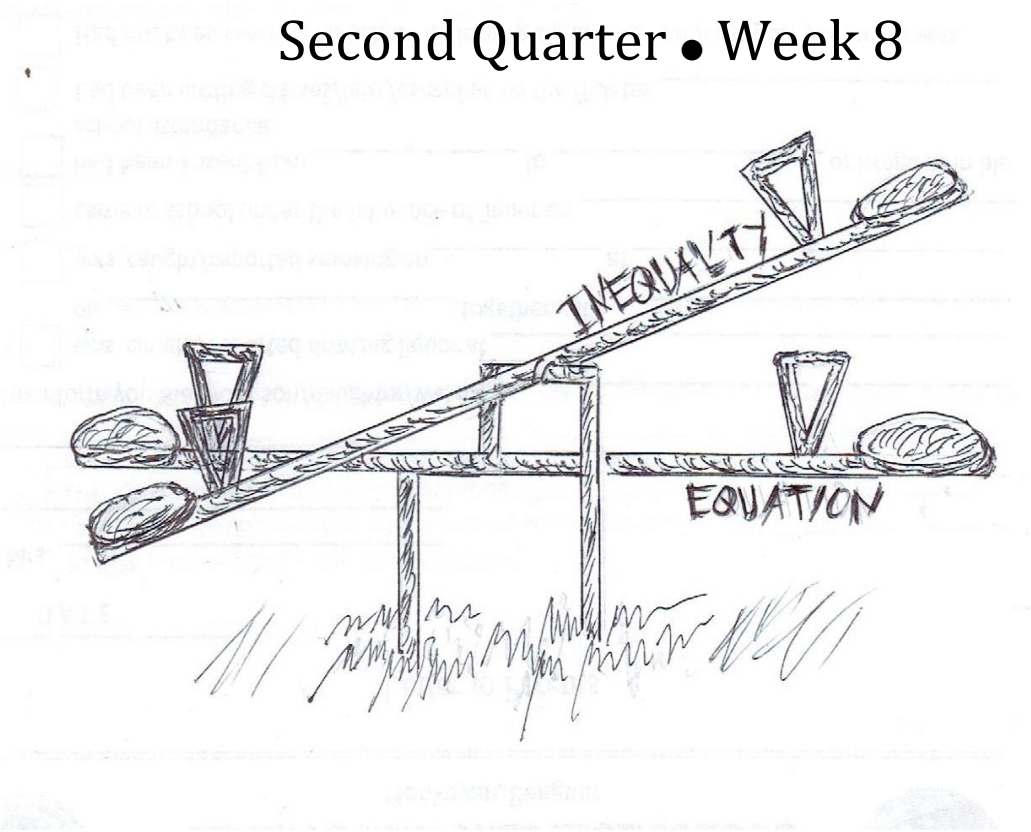
7

Illustrating Linear Equation and Inequality in One Variable

Module in Mathematics 7

Second Quarter • Week 8



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PREFACE

This module is a project of the Curriculum Implementation Division particularly the Learning Resource Management and Development Unit, Department of Education, Schools Division of Benguet which is in response to the implementation of the K to 12 Curriculum.

This Learning Material is a property of the Department of Education- CID, Schools Division of Benguet. It aims to improve students' performance specifically in Mathematics.

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To my family my parents and siblings, whose love and guidance are with me in whatever I pursue. Most importantly, to my kids who provide me with unending inspiration and a reason to do things.

Above all, to our Almighty God for his unending blessing….

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What I Need to Know

In this module, we are going to learn about linear equations and inequality in one variable especially in illustrating. So, at the end of this module, you must be able to learn the given competencies below.

Learning competencies:

1) Differentiates algebraic expressions, equations, and inequalities
2) Identify linear equation and inequality in one variable
3) Illustrate linear equation and inequality in one variable

What I Know

To measure your prior knowledge about the lesson on differentiating equations and equalities, take this pre-test.

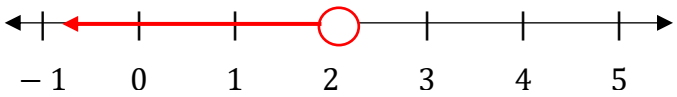
Multiple choice: Choose and write the letter of the correct answer on a separate sheet of paper.

1) It is a statement that two given quantities are unequal.
   A) equation
   B) solution
   C) inequality
   D) problem

2) Which of the following is the sign for "is greater than".
   A) $<$
   B) $>$
   C) $\leq$
   D) $\geq$

3) The sign " =" is used in an _____ means that the two expressions are equal.
   A) equation
   B) solution
   C) inequality
   D) problem

4) Which of the following matches the illustration.



   A) $x \geq 2$
   B) $x > 2$
   C) $x \leq 2$
   D) $x < 2$

5) From the illustration in number 4, what does the unshaded circle mean?
   A) 2 is not a part of the solutions
   B) 2 is a part of the solutions
   C) 2 is only the solution
   D) it cannot be determined

6) Choose the correct description for $x > -1$.
   A) x are all numbers greater than – 1
   B) x are all numbers smaller than – 1
   C) –1 and all numbers smaller than – 1
   D) –1 and all numbers greater than – 1

7) "At most" is clue words for...

A) >

B) <

C) ≤

D) ≥

8) What does this symbol ≤ represent?

A) is less than

B) is greater than

C) is less than or equal to

D) is greater than or equal to

9) Which of the following is true?

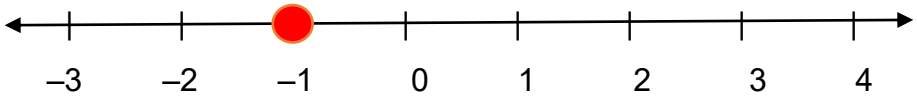
A) 2 + (- 5) = 4

B) 9 + 7 = 4 + 12

C) 8 + 3 > 4 + 7

D) 9 – 7 < 1

10) In the illustration below, what does the shaded circle means?



A) – 1 is the boundary of the solutions

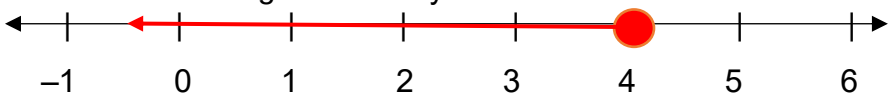
B) – 1 is not a solution

C) – 1 is only the solution

D) it cannot be determined

11) Which of the following illustrates y ≤ 4?

A



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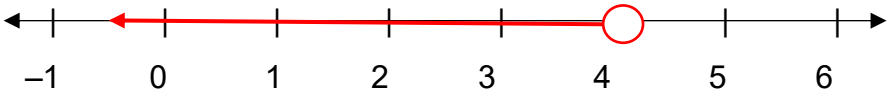

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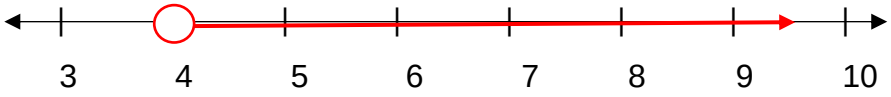

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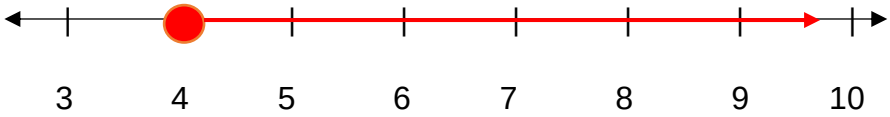
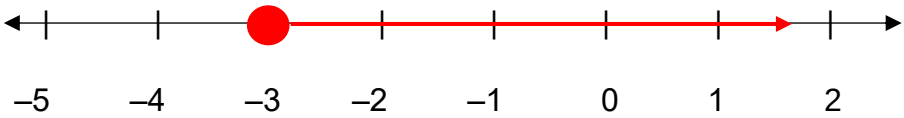


12) Which of the following matches the illustration



A) $z < -3$

B) $z > -3$

C) $z \leq -3$

D) $z \geq -3$

13) In the illustration in number 10 what does the shaded circle means?

A) – 3 is a part of the solutions

B) – 3 is not a part of the solutions

C) – 3 is only the solutions

D) it cannot be determined

14) What does these inequality $x \leq 2$ means?

A) the value of x is from 1, 0, –1, – 2, –3…….

B) the value of x is from 2, 1, 0, –1, – 2, –3…….

C) the value of x is from 1, 2, 3, 4, 5, 6…….

D) the value of x is from 2, 3, 4, 5 6, 7…….

15) From the illustration below, what are the solutions?

A) the solutions starts with 4, 5, 6, 7, 8, 9, 10,……

B) the solutions starts with 5, 6, 7, 8, 9, 10,……

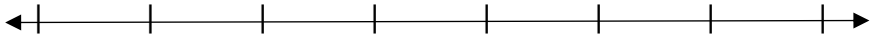
C) the solutions start with 4, 5, 6, 7, 8, 9, 10

D) the solutions start with 5, 6, 7, 8, 9, 10

What's In?

1. Arrange the following in the number line.

A) – 6, – 2, 0, 4, 2, –4, 6



Great!!!! You got it….

In the number line zero (0) is the center, the number in its left side are negative numbers and the numbers in the right side are positive numbers.

What's New

How will you illustrate the given algebraically in the number line?

1) $x + 5 = 9$

2) $y – 2 = – 5$

3) $y + 4 > 2$

4) $x – 8 < – 3$

5) $y + 9 \leq 12$

6) $z + 20 \geq 16$

What's In It?

Before we are going to illustrate the given above. Let's have a discussion on how to differentiate algebraic expressions, equations and inequalities.

Discussion 1:

Algebraic expression is a combination of numbers, variables and can be separated by addition and subtraction.

Ex. 1) $5x + 12$ 2) $9x^3y^2 - 3x^3y^2 + 8$ 5) $5xy^3$

3) $3ab^2c^3 - 7ab^2c^3$ 4) $3a^2b - 2a^2b$

Equation is a Mathematical sentence which states that two expressions are equal. The symbol "=" is used to denote equality.

Ex. 1) $x + 7 = 8$ 3) $8x + 5 = 3x - 10$ 5) $12 - 2(x - 8) = x + 5$

2) $4x - 2 = 5$ 4) $x - 2(10x + 3) + 10 = -5(3x + 4)$

Inequality is a Mathematical sentence showing that the two expressions are not equal. The symbols ">", "< " , " ≥" , " ≤ " indicates inequality.

Ex. 1) $x - 6 < 4$ 3) $4x + 7 < 2x - 5$

2) $3x - 5 \leq x + 4$ 4) $\frac{8x - 3}{3} \leq 5x - 8$

Activity I:

Distinguish the following if it is an **Algebraic Expression, Equation** and **Inequality.**

Inequality 1) $3y + 9 > 4y$

_____2) $4b \leq 3b - 4$

_____3) $x - 8$

_____4) $3c^3d^2 - 6c^3d^2$

_____5) $x^2 + 3x - 6$

_____6) $6t + 5 \geq 2t - 3$

_____7) $x^2 - 8x^2$

_____8) $2p = p - 4$

_____9) $9y > 5y + 8$

_____10) $7x = 21$

Assessment I:

The given are examples of algebraic expression, equation and inequality. To differentiate them try to put the examples which column they belong.

Algebraic Expression	Equation	Inequality

A) $4x + 25$ B) $ab - 4ab + 6cd$ C) $5x + 6 = 9$ D) $9m - 7 > 12$

E) $7x^2y^3z^4 + bc - 8bc$ F) $7w \leq 21$ G) $x - 15 = 12$ H) $x^3 + 5x^2 - 8$

I) $2x^3 - 4x^3 = 18$ J) $b^2 + 4 \geq 20$

Discussion 2:

This time we are going to illustrate linear equation and inequality algebraically and in the number line. These are problem given in "What's New". Let's illustrate them one at a time.

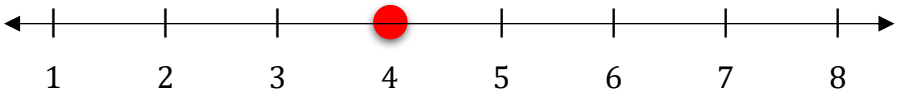
When we illustrate linear equation algebraically, we have

1) $x + 5 = 9$

$x = 9 - 5$

$x = 4$ the value of x is 4

to illustrate $x = 4$ in the number line, we have

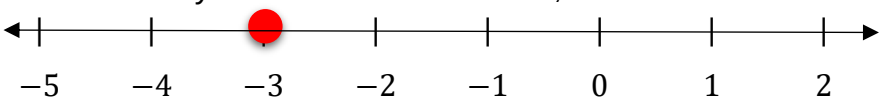


2) $y - 2 = -5$

$y = -5 + 2$

$y = -3$ the value of y is -3

to illustrate also $y = -3$ in the number line, we have



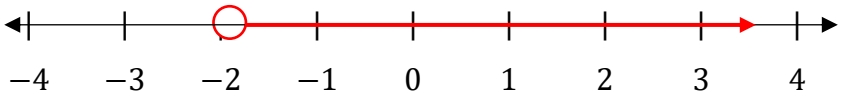
How about an i**nequality**, how do we illustrate them? It's just the same with equation. Let's illustrate them algebraically and also in the number line.

3) $y + 4 > 2$

$y > 2 - 4$

$y > -2$ the value of y is more than negative two,

to illustrating in the number line, we have



When we write all the value of y we start at –1, 0, 1, 2……., the (…..) emphasis that there are still numbers that follows.

Ex: 6,7,8,9 this means that these are only the values, while

3,4,5,6……. means that there are still numbers aside from 3,4,5,6

Take note: The <u>open circle</u> in the illustration shows that <u>– 2 is not a value of y</u>

4) $2x - 8 < 2$

$2x < 2 + 8$

$2x < 10$

$\frac{2x}{2} < \frac{10}{2}$

$x < 5$ the value of x is less than

to illustrate on the number line, we have

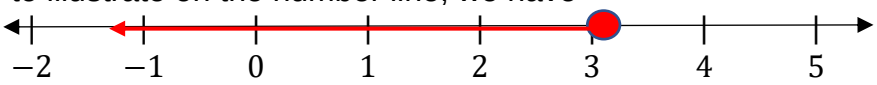
the value of y starts with 4,3,2,1…it shows that 5 is not included

5) $y + 9 \leq 12$

$y \leq 12 - 9$

$y \leq 3$ the value of y is less than or equal to 3

to illustrate on the number line, we have



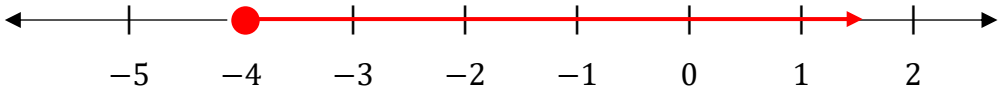
*The value of z starts with 3,2,1,0…

Take note: The <u>shade circle</u> in the illustration shows that <u>3 is included in the solutions</u>

6) $z + 20 \geq 16$

$z \geq 16 - 20$

$z \geq -4$ the value of z is greater than or equal to – 4



*The value of z is –4,–3,–2,–1……

Clue:

The **<u>circle is shaded</u>** when we have the inequality **<u>symbol ≥ and ≤.</u>**

The **<u>circle is unshaded</u>** when we have the inequality **<u>symbol < and >.</u>**

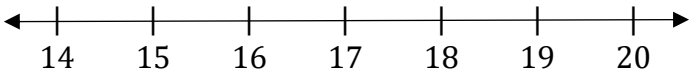
Activity 2:

Try this out: Solve the given problem.

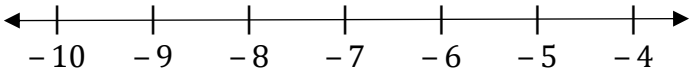
1) $3x + 7 = 28$
2) $x - 5 > -11$
3) $4x + 3 \leq 15$

Illustrate the given problem on the number line.

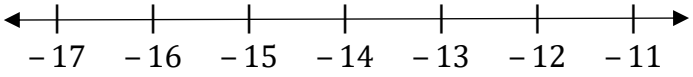
4) $y \leq 18$



5) $y = -9$



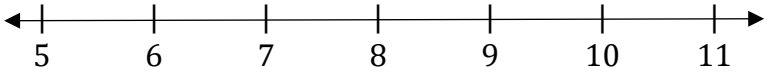
6) $z < -12$



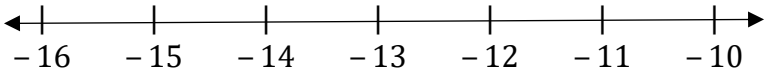
Assessment 2:

Let's try if you really understand how to solve and illustrate linear equation and inequality in one variable. Solve and illustrate the given.

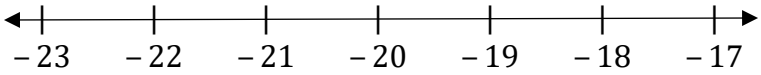
1) $x - 6 > 3$



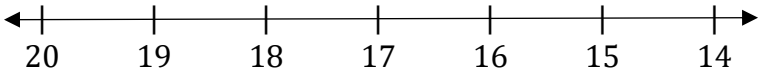
2) $x + 8 \geq -7$



3) $x + 12 = -9$



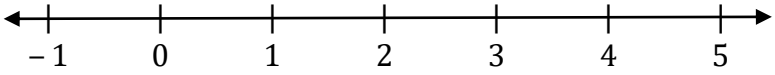
4) $x - 12 < 4$



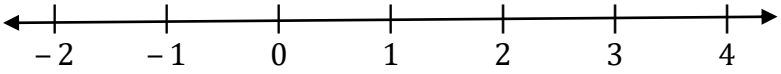
What's More

A. Solve and illustrate the following problems.

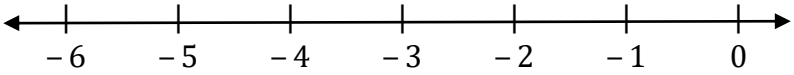
1) $3x - 2 < 10$



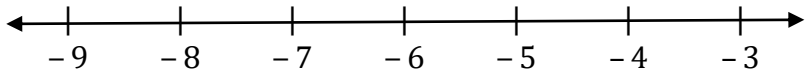
2) $5x - 9 = 5 - 2x$



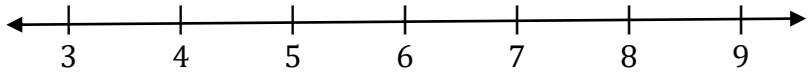
3) $3x - 2 \leq 7x + 10$



4) $7y + 12 > 4y - 9$



5) $3x - 5 \geq 13$



What I Have Learned

A. Differentiate the following terms, algebraic expression, equations and inequalities **<u>in your own word with 2 examples each.</u>**

1) Algebraic Expression _____

_____

Ex: _____

2) Equation _____

_____

Ex: _____

3) Inequalities _____

_____

Ex: _____

4) When do we shade the circle in the number line? _____

5) When do the circle not be shaded? _____

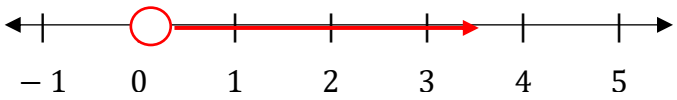
B. <u>Identify</u> and solve linear equation and inequality in one variable then <u>illustrate</u> on the number line.

1) $2x^2 + 7 = 25$
2) $y - 3 < 5$
3) $z^2 + 2x + 1 = 0$
4) $k + 4 = -6$
5) $2x + 9 \geq 19$
6) $3x^3y^2 + x^3y^2 = 8$
7) $2y^3 + 5y^2 \leq 9$

Post Assessment

Multiple choice: Choose and write the letter of the correct answer on a separate sheet of paper.

1) It is a statement whose quantities are unequal.
   A) equation
   B) solution
   C) inequality
   D) problem
2) A combination of numbers, variables separated by addition or subtraction.
   A) $<$
   B) $>$
   C) $\leq$
   D) $\geq$
3) We used the sign " =" for equation showing that
   A) the two quantities are unequal
   B) the two quantities are equal
   C) the two quantities could be unequal or equal
   D) the two quantities neither unequal nor equal
4) Which of the following matches the illustration



   A) $x \geq 0$
   B) $x > 0$
   C) $x \leq 0$
   D) $x < 0$
5) Choose the correct description for $x \geq -5$.
   A) All numbers greater than – 5
   B) All numbers smaller than – 5
   C) – 5 and all numbers smaller than – 5
   D) – 5 and all numbers greater than – 5
6) "At least" means
   A) $>$
   B) $>$
   C) $\geq$
   D) $\leq$
7) What does this symbol $\geq$ represent?
   A) is less than
   B) is greater than
   C) is less than or equal to
   D) is greater than or equal to

8) Which of the following is an example of algebraic expression?

A) $3 \geq 4$

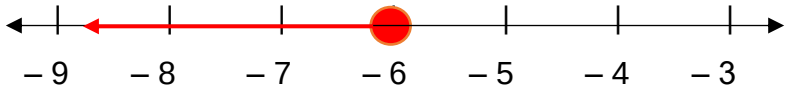
B) $7 = 12$

C) $3x + 7$

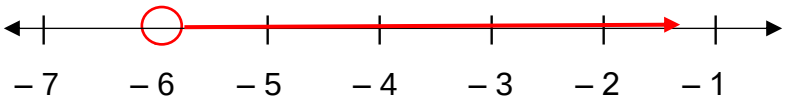
D) $12 < 8$

9) Which of the following illustrates $y < -6$?

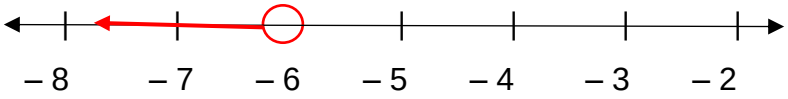
A)



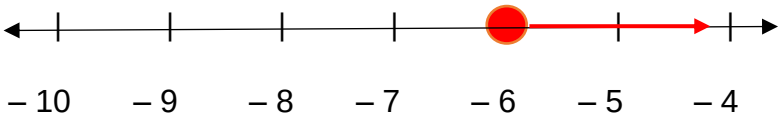
B)



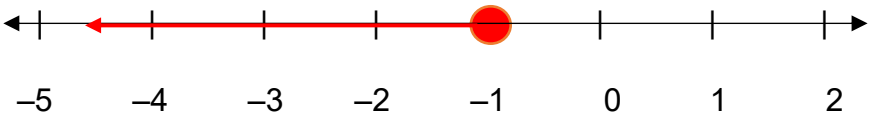
C)



D)



10) Which of the following matches the illustration



A) $z < -1$
B) $z > -1$
C) $z \leq -1$
D) $z \geq -1$

11) If you list down all the solutions in number 10, what are they?

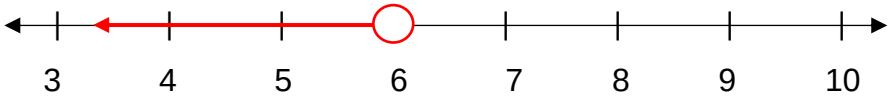
A) – 1, 0, 1, 2…….

B) – 1, – 2, –3, –4,…..

C) 0, 1, 2, 3, 4……..

D) – 2, –3, –4,…..

12) From the illustration below, what are the solutions?



A) the solutions starts with 6, 7, 8, 9, 10,….

B) the solutions starts with 6, 5, 4, 3, 2,….

C) the solutions start with 5, 4, 3, 2, 1,…..

D) the solutions start with 7, 8, 9, 10,……

13) What does the inequality $y \geq -8$ means

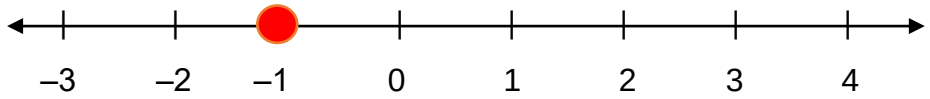
A) the value of y is from –8, – 9, –10, –11…….

B) the value of y is from –8, – 7, – 6, – 5…….

C) the value of y is from –8, – 7,– 6,– 5

D) the value of y is from –8, – 9, –10, – 11

14) Which of the following matches the illustration?



A) $m > -1$

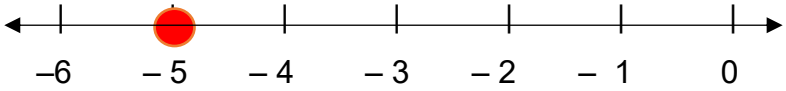
B) $m < -1$

C) $m = -1$

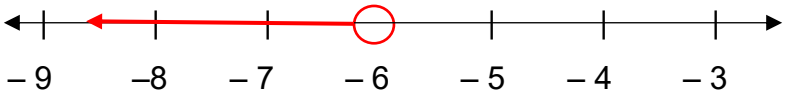
D) $m \geq -1$

15) Which of the following illustration matches $w < -5$

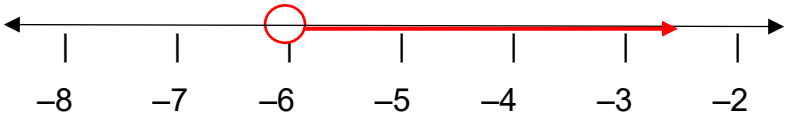
A)



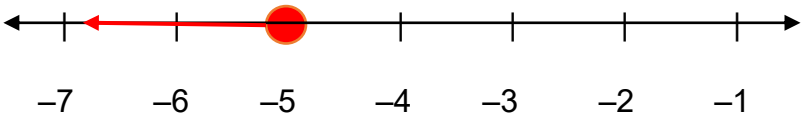
B)



C)



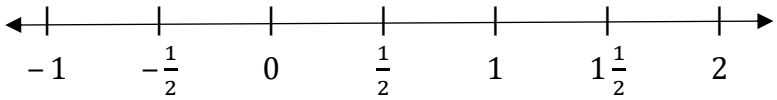
D)



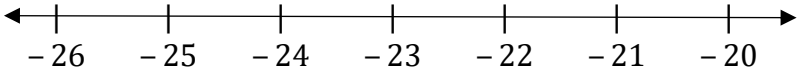
Additional Activity

Solve and ilus the following equation and inequality in one variable using the number line. Provide your own set of values for the given number line.

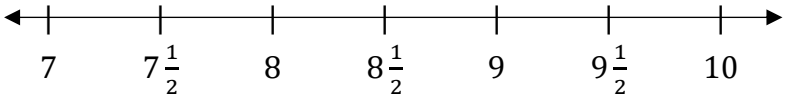
1) $x > \frac{1}{2}$



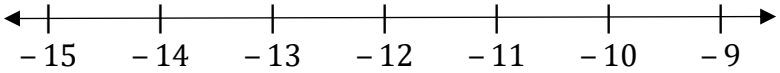
2) $x \geq -25$



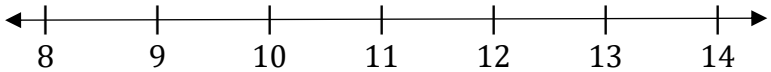
3) x = 9 ½



4) $x \leq -10$



5) x = 11



ANSWER KEY

What I Know:

1) A	6) A	11) A
2) B	7) C	12) D
3) A	8) C	13) A
4) D	9) B	14) B
5) A	10) C	15) B

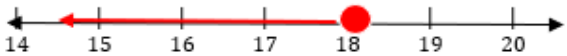
Activity I:

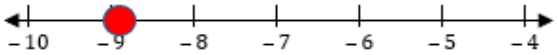
1) Inequality	6) Inequality
2) Inequality	7) Algebraic Expression
3) Algebraic expression	8) Equation
4) Algebraic expression	9) Inequality
5) Algebraic expression	10) Equation

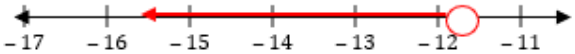
Assessment I:

Algebraic Expression	Equation	Inequality
A) $4x + 25$	C) $5x + 6 = 9$	D) $9m - 7 > 12$
B) $ab - 4ab + 6cd$	G) $x - 15 = 12$	F) $7w \leq 21$
E) $7x^2y^3z^4 + bc - 8bc$	I) $2x^3 - 4x^3 = 18$	J) $b^2 + 4 \geq 20$
H) $x^3 + 5x^2 - 8$		

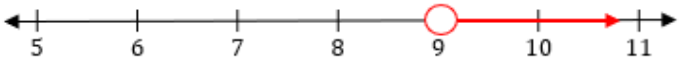
Activity 2:

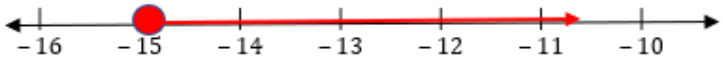
1) $x = 7$
2) $x > -6$
3) $x \leq 3$
4)


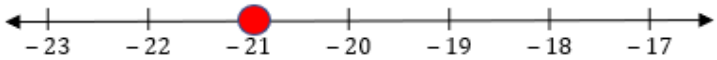
5)


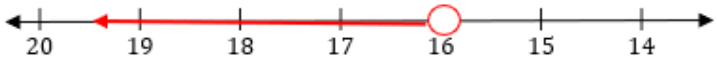
6)


Assessment 2:

1) $x \geq 9$


2) $x \geq -15$


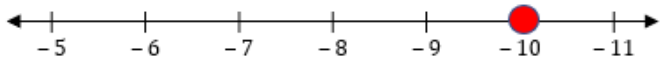
3) $x = -21$


4) $x < 16$


What I Have Learned:

A: 1) 2) 3) Answer may vary
4) When the inequality symbol is ≥ or ≤
5) When the inequality symbol is > or <

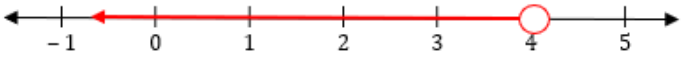
B: 2) $y < 8$

4) $k = 10$


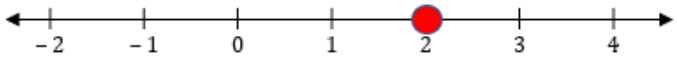
5) $x \geq 5$

What's More:

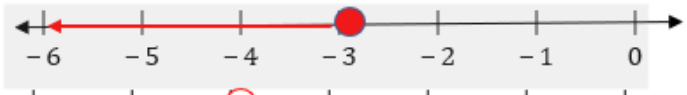
1) $x < 4$



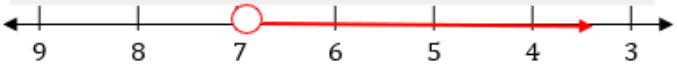
2) $x = 2$



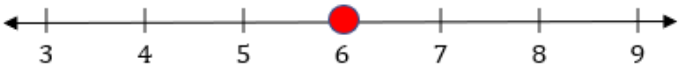
3) $x \leq -3$



4) $y > 7$



5) $x = 6$



What I Can Do

1) a) Tom's age ≥ 12

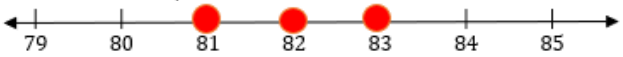
b)

2) a) Kelly save < ₱ 40

b)

3) a) the numbers are 81, 82 and 83

b)



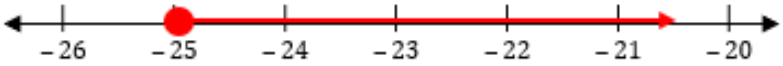
Post Assessment:

1) C	6) C	11) B
2) A	7) D	12) C
3) B	8) A	13) B
4) B	9) C	14) C
5) D	10) C	15) B

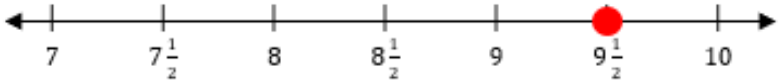
Additional Activity

1) $x > \frac{1}{2}$

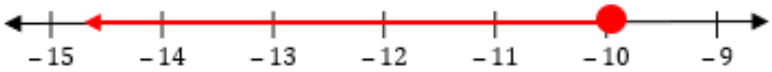
2) $x \geq -25$



3) $x = 9\frac{1}{2}$



4) $x \leq -10$



5) $x = 11$

Reference Sheet

Milagros D. Ibe, Maxima J. Acelajado, Evangeline F. Golla; High School

Mathematics 1 - Concept and operations (Third Edition) Philippine; Copyright 1999 by Diwa Scholastic Press Inc. pp. 336 – 334 pp., 345 – 348

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